The percolation recurrence rate RR_c

A connectedness measure for recurrence plots

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1 The idea

Take a thresholded recurrence plot. When the threshold ε is small, only states that are almost identical count as recurrent, so the plot is just scattered specks. When we raise ε, more pairs of states count as close, the specks grow, start touching each other and merge into bands and blocks. At some point one single connected structure links the *start* of the series to its *end*.

The recurrence rate at which this happens first is what I call the percolation recurrence rate, RR_c. So instead of choosing ε ourselves, ε becomes the output. The question it answers is: how loose does our idea of "similar" have to be before the whole history of the system becomes one chain of similar, connected episodes?

Figure 1 shows this for a simulated series. The share of recurrence points in the largest structure creeps up slowly, and then jumps. Shuffling the same values in time destroys the connectivity completely, which tells us that RR_c is about the time order of the series, not about which values occur.

2 Definition

We start from a series, turn it into states $\vec{x}_1, \ldots, \vec{x}_N$ (delay embedding with dimension m and delay τ), and compute the distances $D_{ij} = \|\vec{x}_i - \vec{x}_j\|$. We also need a Theiler window w and an edge fraction δ (for example $\delta = 0.05$).

- **Domain.** $\Omega = \{(i, j) : j - i \geq w\}$. This is the upper triangle without the main diagonal and the pairs right next to it. Without this, the main diagonal would connect everything trivially.
- **Recurrence points.** $P(\varepsilon) = \{(i, j) \in \Omega : D_{ij} \leq \varepsilon\}$, with recurrence rate $RR(\varepsilon) = |P(\varepsilon)|/|\Omega|$.
- **Connected.** Two recurrence points (i, j) and (i', j') touch if $\max(|i - i'|, |j - j'|) = 1$, so horizontal, vertical and diagonal neighbours (8-neighbourhood). A connected structure is a set of recurrence points that can reach each other through touching points.
- **Start and end.** Let $k = \max(1, \lceil \delta N \rceil)$. Then $A = \{(i, j) \in \Omega : i \leq k\}$ are the first k rows (early times), and $B = \{(i, j) \in \Omega : j > N - k\}$ are the last k columns (late times).

Then

$$\varepsilon_c = \min\big\{\varepsilon : \text{ one connected structure of } P(\varepsilon) \text{ contains points of both } A \text{ and } B\big\},$$
$$RR_c = RR(\varepsilon_c).$$

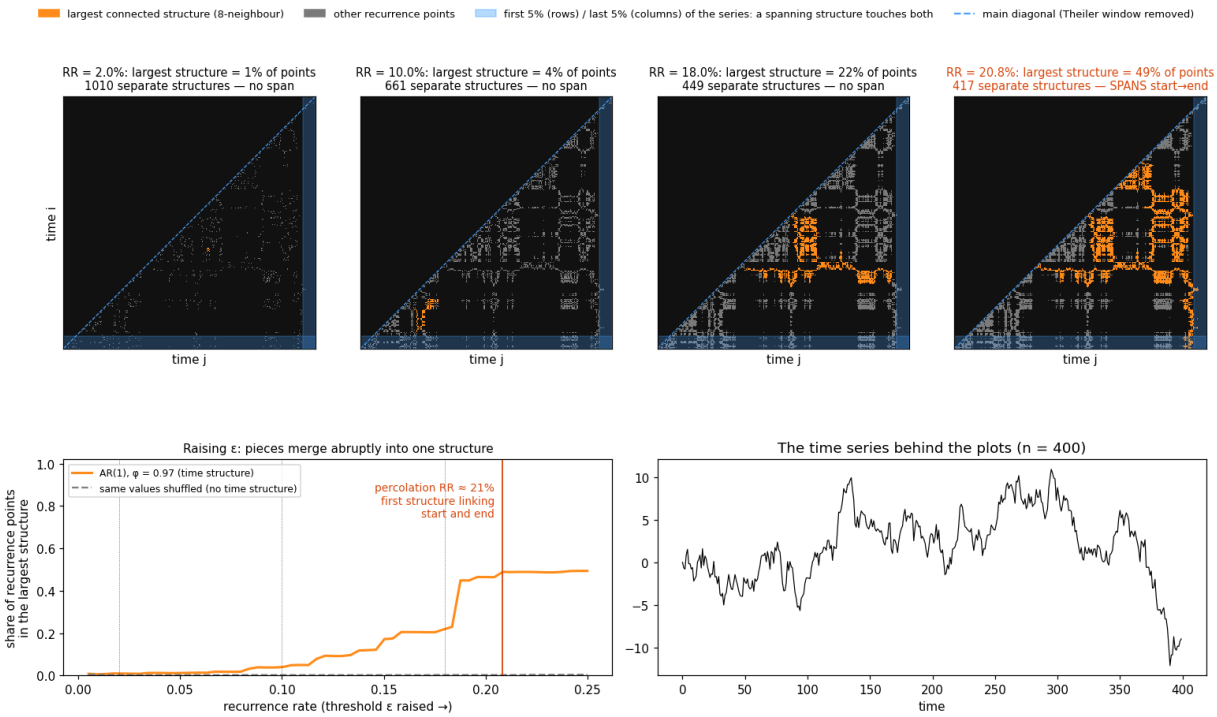


Figure 1: Top: the same recurrence plot of an AR(1) series ($\phi = 0.97$, $n = 400$) at four recurrence rates. Orange is the largest connected structure. The blue bands are the first 5% of rows and the last 5% of columns. At $RR \approx 21\%$ the orange structure touches both, so it spans. Bottom left: share of recurrence points in the largest structure, for the series and for a shuffled copy.

RR_c lies between 0 and 1. A low value means the history links up early, at a fairly strict notion of similarity. A high value means the history breaks into separate episodes that only connect once almost everything counts as similar.

Example. For the monthly US unemployment rate from 1954-07 to 2026-09 we have $N = 867$ months. With $\delta = 0.05$ we get $k = \lceil 43.35 \rceil = 44$, so A is the first 44 months (1954-07 to 1958-02) and B is the last 44 months (2023-02 to 2026-09). RR_c is then the recurrence rate at which a single structure first contains both a late-1950s month and a 2023–26 month.

3 How it is computed

We increase the threshold through the distinct observed distances. At each threshold, all pairs with that distance are added together, so ties are handled consistently. A union–find structure keeps track of connected components and whether each component touches the start and end regions.

Algorithm 1 Percolation recurrence rate

Require: Distance matrix D; Theiler window $1 \le w < N$; edge fraction $0 < \delta < 1/2$
Ensure: Critical threshold ε_c and recurrence rate RR_c

$\Omega \leftarrow \{(i, j) : 1 \le i, j \le N,\ j - i \ge w\}$
$k \leftarrow \max(1, \lceil \delta N \rceil)$
$A \leftarrow \{(i, j) \in \Omega : i \le k\}$; $B \leftarrow \{(i, j) \in \Omega : j > N - k\}$
Sort the distinct values of $\{D_{ij} : (i, j) \in \Omega\}$ increasingly
$P \leftarrow \varnothing$; initialise an empty union–find structure
for each threshold d in the sorted values **do**
    $Q \leftarrow \{(i, j) \in \Omega : D_{ij} = d\}$
    for each pixel $p \in Q$ **do**
        Add p to P as a singleton component
        Set its flags to $\mathbf{1}[p \in A]$ and $\mathbf{1}[p \in B]$
    end for
    for each pixel $p \in Q$ **do**
        for each active 8-neighbour $q \in P$ of p **do**
            Merge their components, combining each flag by logical OR
        end for
    end for
    if a component containing a pixel of Q has both flags **then**
        return $\varepsilon_c = d$, $RR_c = |P|/|\Omega|$
    end if
end for

For $M = |\Omega|$, sorting takes $O(M \log M)$ time. Each pixel has at most eight neighbours, so component updates take $O(M\alpha(M))$ time, where α is the inverse Ackermann function and grows extremely slowly. Memory use is $O(M)$. Since $M = O(N^2)$, the overall time is $O(N^2 \log N)$.

4 What to report next to it

On its own, RR_c will mostly be low for smooth series, just because neighbouring months are similar. So it should always be compared with a surrogate: a series with the same autocorrelation but no other structure (for example a fitted AR model, or an IAAFT surrogate). The statistic we actually care about is the difference $RR_c - RR_c^{\text{surr}}$, or where the real RR_c falls among many surrogates. We also have to state the choices we made: m, τ, w, δ, the 8-neighbourhood, and the distance (Euclidean or a β-divergence).

5 Where it could be useful

These are ideas to test, not results yet.

- **Regimes and breaks.** A series with distinct regimes, or a one-off break, should only link up late, so its RR_c is high. In the example in Figure 1, the final drop to -10 delays the

span, because the last months look like nothing that came before. For macro series like unemployment this could say whether the post-COVID period is really something new or just another version of earlier episodes.

- **No threshold to choose.** Most RQA measures depend strongly on ε. RR_c turns the threshold into the measurement itself, so two series can be compared without picking an ε for each.
- **Rolling windows.** Computed on moving windows, a rising RR_c could mean the system is moving into a state it has not visited before, which is a possible early-warning signal.
- **Joint recurrence.** The same procedure works on a joint recurrence plot. A pair enters the joint plot only once it is recurrent in every variable, so we can rank each pair's distance within each variable, take the largest rank, and run the same procedure on that. Then RR_c asks at what level of similarity the shared history of two or three systems links up, and comparing it with the single-system values shows what the joint plot adds.
- **Geometry.** Computing RR_c with Euclidean distance and with a β-divergence on the same data shows directly how the choice of geometry changes which parts of the history count as connected.
- **Synthetic data checks.** Generated market data should link up like the real data does. A very different RR_c would mean the generator gets the time structure wrong.

6 Limits

- The span rule depends on what happens at the edges of the series, and on δ. A different version is to take the recurrence rate where the largest-structure share jumps the most, which needs no edge regions.
- Longer series and finer sampling make spanning easier, so values are only comparable at the same length and sampling.
- 8-neighbourhood joins diagonal lines but can also merge strands that have nothing to do with each other. 4-neighbourhood does the opposite.

7 Related work

Percolation and connectivity have been studied for *recurrence networks*, where the nodes are the time points and the edges are recurrences [1, 2]. That is a different object: shuffling the time labels does not change those networks, while it destroys the structures here. Persistent homology of the distance-matrix image tracks how many components appear and merge as ε grows [3], and enclosed holes in recurrence plots have been counted before [4]. Diagonal lines that are bent by time rescaling are described in [5]. In our literature search I did not find a fixed definition of the recurrence rate at which pixel structures first span the plot.

References

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